Contextualizing Fairness: Extending p-Mean Social Welfare to Linear Bandits

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Course: CS6007:Multi-Agent Machine Learning

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October 31, 2026

Outline

1. Problem Statement and Motivation
2. Welfarist-UCB: Algorithm and Theory
3. Reproduction of the Paper's Results
4. Extension: p-Mean Regret in Contextual Linear Bandits
5. Conclusion

Setting: Stochastic Multi-Armed Bandits

- k arms with unknown means $\mu_1, \ldots, \mu_k \geq 0$; rewards are σ^2-sub-Gaussian. Let $\mu^\star = \max_i \mu_i$.
- Over T rounds the learner pulls I_t using only past observations. **Each round is a distinct user** (e.g. a patient in a clinical trial).
- Classical objective, the average regret

$$AR_T \;=\; \mu^\star - \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu_{I_t}].$$

- It maximises *total* utility, but it is blind to how rewards are spread over users: a policy can starve a few users and still have small AR_T.
- $\mu_i \geq 0$ is needed for the welfare functions to make sense (arms are "pre-screened" to be harmless on average); individual realised rewards may still be negative.

Fairness Objective: Nash and p-Mean Regret

Welfare of the expected-reward sequence is a *power mean*; regret is the gap to $\mu^\star$:

$$NR_T = \mu^\star - \Big(\prod_{t=1}^{T} \mathbb{E}[\mu_{I_t}]\Big)^{1/T}, \qquad R_T^p = \mu^\star - \Big(\frac{1}{T}\sum_{t=1}^{T} (\mathbb{E}[\mu_{I_t}])^p\Big)^{1/p}.$$

- $p = 1$: utilitarian, $R_T^1 = AR_T$
- $p = 0$: Nash social welfare
- $p \to -\infty$: Rawlsian (worst-off user)
- $p > 1$: favours well-off users

The power mean increases in p:

$$R_T^p \uparrow \text{ as } p \downarrow, \qquad NR_T \geq AR_T.$$

Smaller p weights low-reward rounds more: **more fairness, a harder learning problem**. Region of interest: $p \leq 1$ [4, 6].

Why Is It Hard? What Prior Work Needed

- A single round with near-zero expected reward collapses the geometric mean. **Plain UCB fails**: two Bernoulli arms with $\mu_1 = (2e)^{-T},\ \mu_2 = 1$ [2].
- **NCB** (Barman et al. 2023): index with a width that depends on $\widehat{\mu}_i$, analysed with *multiplicative* Chernoff bounds $\Rightarrow$ rewards bounded and non-negative, a known bound on $\mu^\star$.
- **Explore-then-UCB** (Krishna et al. 2025) for p-mean regret: needs bounded rewards and a minimum mean $\mu_i \geq k\,T^{-1/4}$ for *every* arm.

Question

Can a short *adaptive* uniform exploration phase followed by plain UCB give (near-)optimal Nash and p-mean regret for *all* p, for general sub-Gaussian rewards and with no assumption on the means?

The answer of [6]: **yes**, via Welfarist-UCB.

Contributions and Table of Bounds

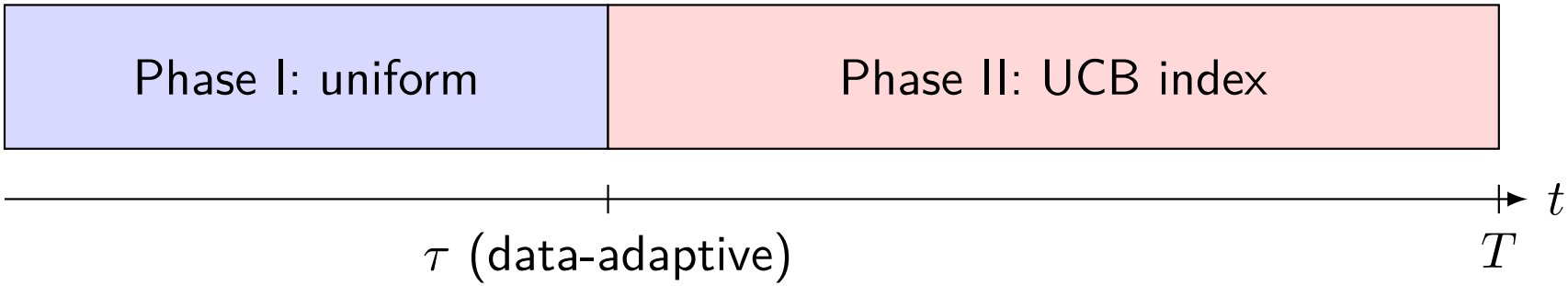
	$p \in [0,1]$	$p \in [-1,0)$	$p < -1$
Barman et al. 2023	$O(\sqrt{k/T})$ at $p=0$	–	–
Krishna et al. 2025	$O(\sqrt{k/T})$	$O(k^{3/4}T^{-1/4})$	$O(k^{1/2}T^{-1/(4\lvert p\rvert)})$
Welfarist-UCB	$O(\sqrt{k/T})$	$O(k/\sqrt{T})$	$O(\lvert p\rvert\, k^{\frac{\lvert p\rvert+1}{2}}/\sqrt{T})$

(up to $\sqrt{\log T}$ factors)

Assumptions: Barman/Krishna need rewards in $[0,1]$ (and Krishna $\mu_i \geq kT^{-1/4}$); Welfarist-UCB needs only **sub-Gaussian rewards and** $\mu_i \geq 0$.

Uses the *additive* Hoeffding bound only, so it covers Gaussian rewards and needs no bound on $\mu^\star$.

Welfarist-UCB: Two Phases



Phase I (until the stopping rule fires)

- every block of k rounds, draw a uniform random permutation of the arms and pull them in order
- equivalent to uniform sampling: $\mathbb{E}[\mu_{I_t}] \geq \mu^\star/k$ per round
- track n_i and $\widehat{\mu}_i$

Phase II

- pull $I_t = \arg\max_i \ \widehat{\mu}_i + 2\sqrt{\frac{2\sigma^2 \log T}{n_i}}$
- the *standard* UCB index, no estimate-dependent width (no NCB)

The whole difficulty is **when to stop Phase I**: stop early and UCB may pick a near-zero arm; stop late and uniform exploration wastes welfare.

The Data-Adaptive Stopping Rule

With $w_i = 2\sqrt{2\sigma^2 \log T / n_i}$ (UCB width), Phase I ends at the first t at which, for some arm i,

$$\widehat{\mu}_i > w_i \quad \text{and} \quad n_i > \frac{192\, p_a^2\, \sigma^2 \log T}{\left(\widehat{\mu}_i - w_i\right)^2}, \qquad p_a = \begin{cases} 1 & p \geq -1 \\ |p| & p < -1 \end{cases}$$

- With $\mathrm{LCB}_i = \widehat{\mu}_i - w_i$ the rule reads $\mathrm{LCB}_i > 0$ and $\mathrm{LCB}_i \geq \kappa\, p_a\, w_i$ for a constant $\kappa \approx 5$ ($\kappa = 10$ in the reference code): *the best arm is certified to be much larger than its own confidence width*.
- It uses the empirical mean $\widehat{\mu}_i$ in place of the unknown $\mu^\star$: the rule is **data-driven**.
- For $p \geq -1$ it does not depend on p; for $p < -1$ it scales with $p^2 \Rightarrow$ longer exploration.

Why This Rule Makes UCB Safe

Let $S = 4p_a^2\sigma^2 \log T/(\mu^\star)^2$.

Lemma (length of Phase I)

On the good event, $32\,kS \le \tau \le 128\,kS$, i.e. $\tau = \Theta\big(k\,p_a^2\sigma^2 \log T/(\mu^\star)^2\big)$.

- Before $32kS$ no arm satisfies the condition; by $128kS$ the optimal arm does (its $n_i\,(\mu_i - w_i)^2$ is large).
- Hence every arm has $n_i \gtrsim 1/(\mu^\star)^2$ at the end of Phase I, so the UCB width is a *small constant fraction of* $\mu^\star$.
- Then any arm pulled in Phase II has $\mu_i \ge \mu^\star - 4\sqrt{2\sigma^2 \log T/(T_i - 1)}$, which is close to $\mu^\star$, so its contribution to the geometric / power mean is small.
- Cost of Phase I: each round earns $\ge \mu^\star/k$ in expectation, so its factor in the geometric mean is $k^{-\tau/T} \ge 1 - \tau \log k/T$.

Main Theorems

Theorem 1 (Nash regret)

For any k-armed instance with σ^2-sub-Gaussian rewards and moderately large T,

$$NR_T \;=\; O\Big(\sqrt{\frac{k \log T \, \log k}{T}}\Big).$$

Theorem 2 (p-mean regret)

For fairness parameter $p \in \mathbb{R}$,

$$R_T^p = \begin{cases} O\Big(\sqrt{\frac{k \log T \, \log k}{T}}\Big), & p \geq 0, \\ O\Big(k^{\frac{|p|+1}{2}} \sqrt{\frac{\log T}{T}} \; \max\{1, |p|\}\Big), & p < 0. \end{cases}$$

Standard σ is treated as a constant. For $p \in [-1, 0)$ the bound is $O(k/\sqrt{T})$.

Remarks on the Theorems

- **Optimality.** For $p \leq 1$, $R_T^p \geq AR_T = \Omega(\sqrt{k/T})$. So the bound is optimal up to polylog factors for $p \in [0, 1]$ and off by at most a factor k for $p \in [-1, 0)$.
- **vs. NCB and Explore-then-UCB.** Strict generalisation of the Nash bound of NCB (no boundedness, no bound on $\mu^\star$); for $p \in [-1, 0)$, $O(k/\sqrt{T})$ beats $O(k^{3/4}T^{-1/4})$ and no minimum mean is needed.
- **No free fairness.** For $p < -1$ the bound grows exponentially in $|p|$ and is non-vacuous only when $T \gtrsim p^2 k^{|p|}$; as $p \to -\infty$ vanishing Rawlsian regret is impossible in finite time. The T-dependence stays $1/\sqrt{T}$; the intrinsic hardness grows.
- **Open.** A matching lower bound for $p < -1$.

Proof Idea

Condition on the good event $\mathcal{E}$: $|\widehat{\mu}_i - \mu_i| \le 2\sqrt{2\sigma^2 \log T / n_i}$ for all i, t (additive Hoeffding).

1. **Phase I length** $\tau = \Theta(k p_a^2 \sigma^2 \log T / (\mu^\star)^2)$ from the stopping rule.
2. **Near-optimality of Phase-II arms:** an arm pulled for the T_i-th time had the largest UCB, so $\mu_i \ge \mu^\star - 4\sqrt{2\sigma^2 \log T / (T_i - 1)}$.
3. **Welfare of the two phases separately:** Phase I $\ge (\mu^\star / k)^{\tau/T}$ (uniform sampling); Phase II via multivariate Jensen with $(1-x)^a \ge 1 - 2ax$ and Cauchy–Schwarz over arms.
4. $p \ge 0$**:** power-mean monotonicity gives $R_T^p \le N R_T$. $p < 0$**:** set $q = -p$, split $\sum_t (\mathbb{E}\mu_{I_t})^{-q}$ over the phases and linearise; the p_a^2 in τ produces $k^{(|p|+1)/2}$.

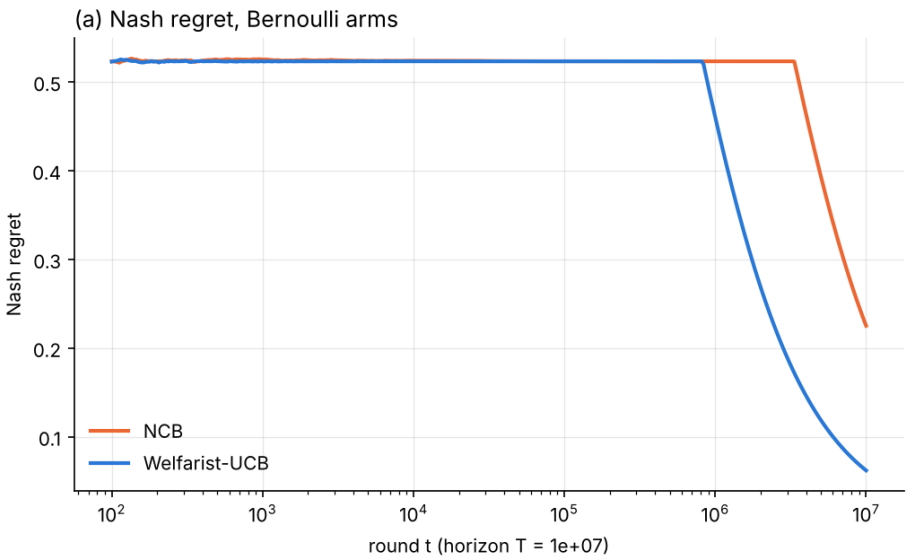
Take-away: the stopping rule *reduces* welfare-regret minimisation to average-regret minimisation (plain UCB) after a short adaptive warm-up.

Reproduction Setup

	instance	T	p	algorithms
(a)	k=50 Bernoulli, $\mu \sim U[0.005, 1]$	10^7	0	NCB, W-UCB
(b)	k=50 Gaussian, $\mu \sim U[10, 1000]$, σ=20	10^6	0	NCB, W-UCB
(c)–(e)	same Gaussian setup, new draw	10^6	$0.5, -0.5, -1.5$	Explore-then-UCB, W-UCB
(f)	same Gaussian setup, new draw	10^6	$1, 0, -1, -2, -5$	W-UCB

- W-UCB = Welfarist-UCB. 50 runs each; instances drawn as in the authors' reference code.
- Plots show $\mu^\star - M_p(\mathbb{E}[\mu_{I_1}], \ldots, \mathbb{E}[\mu_{I_t}])$ over the first t rounds of a horizon-T run.
- Constants follow the reference code (width $2\sqrt{2\sigma^2 \log T / n}$; stopping constant $800\, p_a^2 \sigma^2 \log T$ in place of the paper's 192).
- Baselines: NCB [2] and Explore-then-UCB [3].

Result 1: Nash Regret, Welfarist-UCB vs. NCB

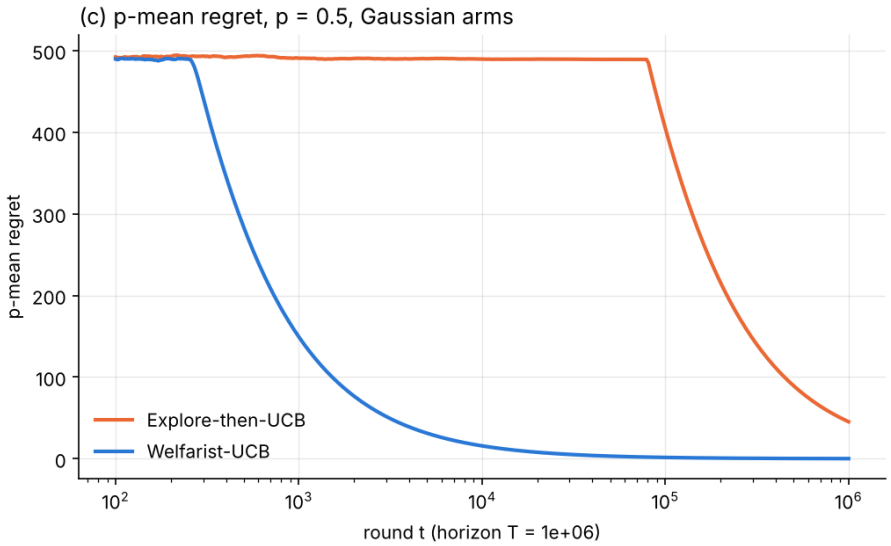


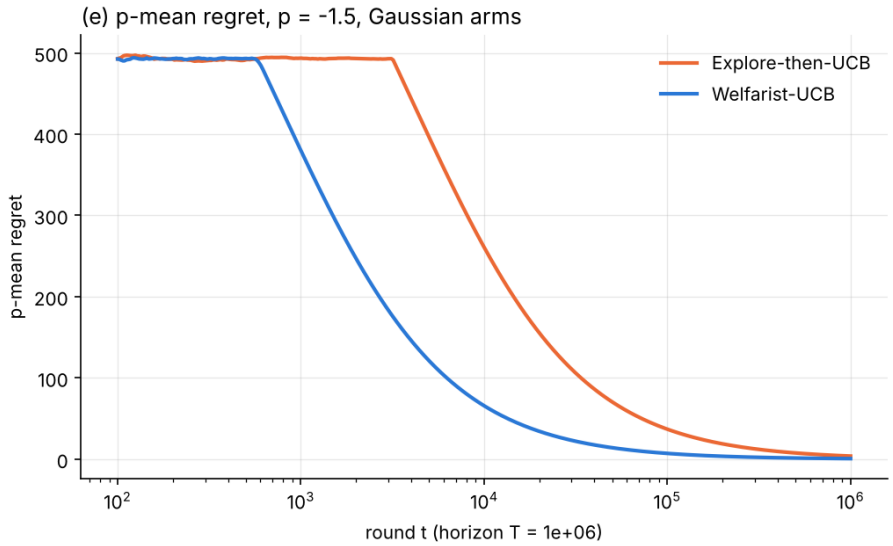
(a), $T{=}10^7$	NCB	W-UCB
Nash regret	0.225	**0.062**
Phase I (10^6 rounds)	3.35	0.83

(b), $T{=}10^6$	NCB	W-UCB
Nash regret	1.75	**0.18**
Phase I (rounds)	2487	266

- Reproduces the trend of Fig. 1(a) in the paper: the adaptive rule stops exploring about $4\times$ sooner, so Welfarist-UCB's regret starts falling earlier and ends at 0.062 vs. 0.225.

Result 2: p-Mean Regret vs. Explore-then-UCB

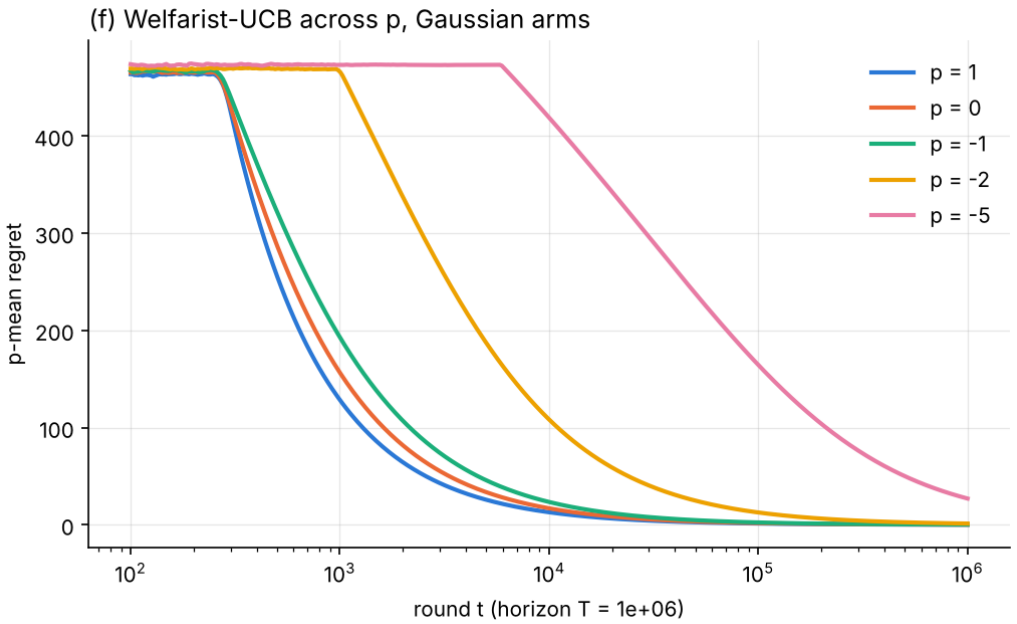




regret at $T = 10^6$	$p = 0.5$	$p = -0.5$	$p = -1.5$
Explore-then-UCB	45.3	18.1	3.85
Welfarist-UCB	**0.16**	**0.22**	**0.72**

Welfarist-UCB wins at every p, by two orders of magnitude for $p \geq -0.5$.

Result 3: Effect of the Fairness Parameter p



p	regret at T	Phase I
1	0.13	276
0	0.17	273
-1	0.24	275
-2	1.29	1010
-5	27.0	5922

- **Observation:** regret grows as p decreases. For $p \geq -1$ Phase I is about 275 rounds, independent of p; for $p < -1$ the stopping rule scales with p^2 and exploration grows $3.7\times$ $(p = -2)$ and $21\times$ $(p = -5)$: the "no free fairness" phenomenon, visible in practice.

Reproduction: Observations

- **The paper's qualitative findings reproduce** (Fig. 1 a–f): Nash regret 0.062 (Welfarist-UCB) vs. 0.225 (NCB) at $T = 10^7$; Welfarist-UCB beats Explore-then-UCB at every tested p.
- **The adaptive stopping rule is the source of the gain**: ~ 270 exploration rounds for $\sigma = 20$ Gaussian arms, against $2{,}487$ (NCB) and $3{,}163$–$79{,}955$ (Explore-then-UCB), with no knowledge of $\mu^\star$.
- **Cost of fairness**: exploration and regret grow with $|p|$ for $p < -1$, exactly as the p^2 factor in the stopping rule and the $k^{(|p|+1)/2}$ factor in the bound predict.
- **Phase I welfare matters**: all the regret before τ is the price of uniform exploration, so the curves are flat until the stop and then fall quickly.

Starting point for the extension: what should the stopping rule look like when arms share features?

What Already Exists

Setting	Nash ($p = 0$)	p-mean (all p)
k fixed arms, no features	Barman et al. 2023 [2]	Krishna et al. 2025 [3], Sarkar et al. 2025 [6]
Linear, fixed arm set	LinNash, Sawarni et al. 2023 [7]: $\tilde{O}(d^{5/4}/\sqrt{T})$	**Sarkar et al. 2026** [5], FairLinBandit: $\tilde{O}(d/\sqrt{T})$
Contextual (changing candidates, user groups)	—	— (our empirical study)

- So the question is *not* whether Welfarist-UCB can be moved to linear bandits: for a fixed arm set, [5] did it with guarantees.
- Our project: carry the *stopping rule idea* to **contextual** linear bandits, where the candidate set changes every round and some feature directions (minority users) are rare, and study it against LinNash and FairLinUCB.

Contextual Setting and Regret

- Round t: user u_t arrives with K candidate items; showing item i gives reward with mean $\langle x_{t,i}, \theta^* \rangle$, where $x = z_u \otimes y_i$ (user features $\otimes$ item features), $d = d_z d_y$. A fixed arm set is the special case of one user.
- The best reward now depends on the context, $r_t^* = \max_i \langle x_{t,i}, \theta^* \rangle$, so

$$R_T^p \;=\; M_p(r_1^*, \ldots, r_T^*) - M_p\big(\mathbb{E}[r_1], \ldots, \mathbb{E}[r_T]\big),$$

  which equals the paper's regret when there are no contexts.
- **Group fairness.** Utility of a group = expected reward its members received / best they could have received; we also report the p-mean of the group utilities.
- Assumptions as in the paper: $\langle x, \theta^* \rangle \geq 0$, σ^2-sub-Gaussian rewards, ridge regression ($\lambda = 1$) with radius $\beta = \sigma\sqrt{2 \log T} + \|\theta^*\|$.

How We Adapt the Algorithm: Explore-then-LinUCB

Keep the two-phase structure, replace UCB by **LinUCB**, and modify the stopping rule.

1. **Phase I sampling.** With prob. $\frac{1}{2}$ show a uniformly random candidate (keeps $\mathbb{E}[r_t] \geq \frac{1}{2}$ of the candidates' average); otherwise the candidate with the largest $\|x\|_{V^{-1}}$, a greedy D-optimal-design step aimed at the least-explored (rare) directions.
2. **Stopping rule** (next slide): certify *all plausible* items, not just the best one.
3. **Groups.** Each protected group gets its own stopping time: a round explores while any group of its user is not yet certified. (Variant: one global stop.)
4. **Phase II: LinUCB**, warm-started with V and $\hat{\theta}$ from Phase I.

Naive transfer (baseline): uniform sampling, the paper's best-arm rule on the current candidates, one global stop, then LinUCB.

The Modified Stopping Rule

Welfarist-UCB: $\exists\, i:\ \mathrm{LCB}_i \ \geq\ \kappa\, p_a\, w_i$, enough because round-robin gives every arm the *same* width. With features, widths $w_x = \beta\|x\|_{V^{-1}}$ **differ by direction**: an item along a rarely explored direction can look plausible long after the common directions are pinned down.

Coverage rule (ours)

Stop when $\mathrm{LCB}^* = \max_a \mathrm{LCB}(x_a) > 0$ and

$$\mathrm{LCB}^* \ \geq\ \kappa\, |p_a|\, \beta\, \|x_a\|_{V^{-1}} \quad \text{for every } a \text{ with } \mathrm{UCB}(x_a) \geq \mathrm{LCB}^*.$$

Items already clearly worse than LCB^* need no further exploration.

- $\kappa = 10$ is the paper's constant; tested on each round's candidates, one stopping time per group.
- Needs design-based sampling: with uniform sampling the rare directions get 1% of pulls and the rule never fires within T.

Relation to FairLinBandit (Sarkar et al., 2026)

	FairLinBandit / FairLinUCB [5]	Ours: Explore-then-LinUCB
Setting	fixed compact arm set $\mathcal{X} \subset \mathbb{R}^d$	changing candidates, user groups
Phase I sampling	$\frac{1}{2}$ D-optimal round-robin, $\frac{1}{2}$ John-ellipsoid distribution	$\frac{1}{2}$ uniform, $\frac{1}{2}$ max-width item
Phase I welfare	$\mathbb{E}[r_t] \geq \mu^\star/(2(d+1))$	$\geq \frac{1}{2}$ candidates' average (no link to r_t^*)
Stopping rule	global, data-adaptive, checked at doubling epochs (below)	plausible-set LCB rule, per group
Phase II	LinUCB or LinPE	LinUCB
Guarantee	$O(d\sqrt{\log T/T})$ for $p \geq 0$; $\times\, d^{\lvert p\rvert/2} \max\{1, \lvert p\rvert\}$ for $p < 0$	none (empirical)

FairLinBandit stops Phase I at the first checkpoint t (starting at $72 \log T$, doubling) with

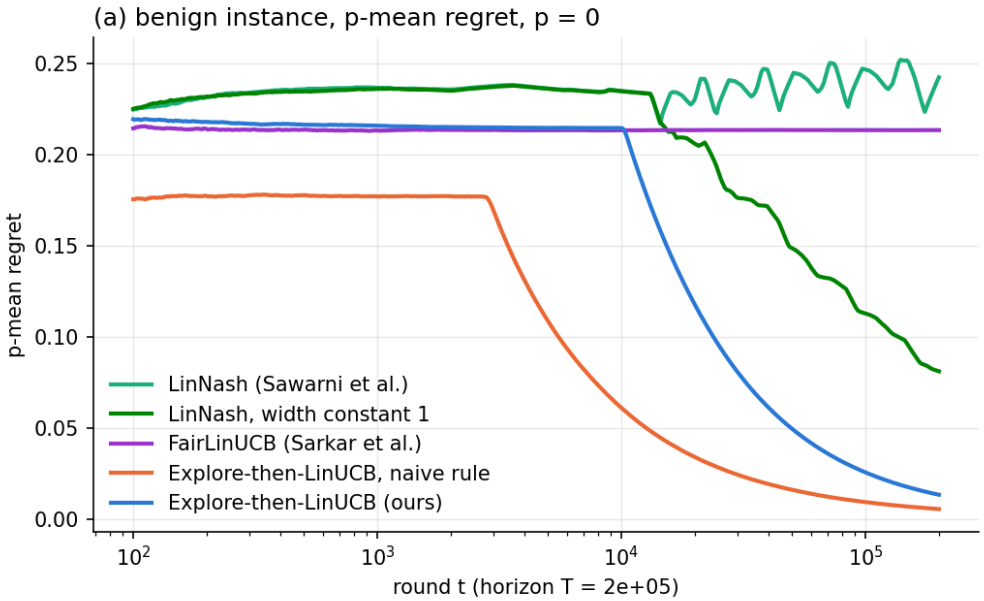
$$t \;\geq\; \max\left\{ \frac{48\sigma^2 d^2 \log T}{\hat{\mu}_{\max}^2},\; \frac{900\, p_a^2 \sigma^2 d^2 \log T}{\left(\hat{\mu}_{\max} - \sqrt{48\sigma^2 d^2 \log T/t}\right)^2} \right\}, \qquad \hat{\mu}_{\max} = \max_x \langle x, \hat{\theta} \rangle.$$

Extension Experiments: Setup

- **(a) Benign fixed-arm**: $K = 50$, $d = 6$, means 0.53–0.97, $\sigma^2 = 0.25$, $T = 2 \cdot 10^5$, 50 runs.
- **(b) Rare direction**: $d = 6$, $K = 400$; only 4 arms (1%) use 2 rare dimensions and the optimum (0.95) is one of them; best common arm 0.85.
- **(c) Two groups**: minority owns a feature block, non-zero only in its rounds, and prefers different items ($d = 8$, $T = 10^5$).
- **MovieLens-1M**: reward = "user likes the movie"; user features one-hot gender, age, occupation; $d = 180$; 20 candidates per round, $T = 5 \cdot 10^4$.

Compared: LinNash [7] (also with width constant 1), FairLinUCB [5] (D-optimal + MVEE-centre, our reimplementation), naive transfer, and ours. LinNash and FairLinUCB need a fixed arm set, so they appear only in (a), (b). **Metric:** p-mean regret, $p \in \{1, 0, -1, -2, -5\}$; the headline comparison with LinNash is $p = 0$.

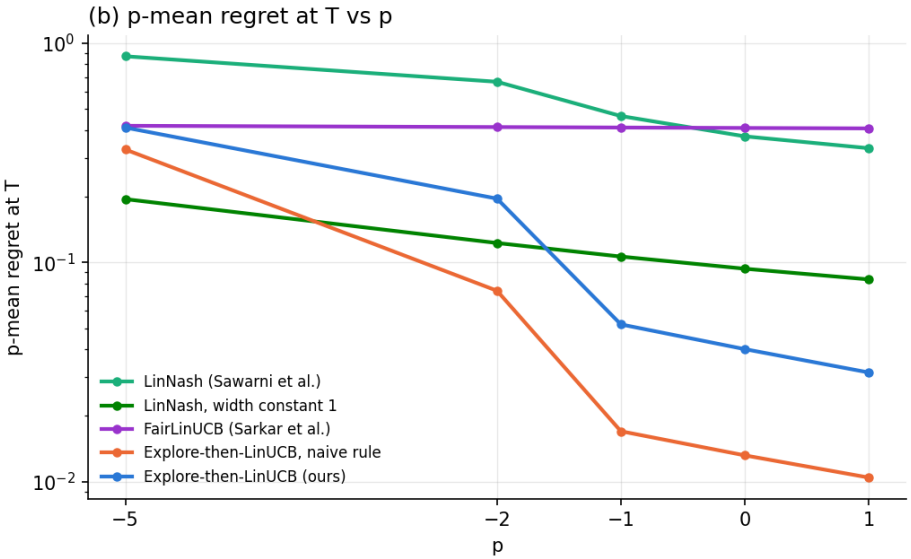
Results at $p = 0$: Nash Regret vs. LinNash



Nash regret at T	(a)	(b)
LinNash (paper const.)	0.243	0.376
LinNash (width const. 1)	0.081	0.094
FairLinUCB	0.214	0.410
Naive transfer	**0.005**	**0.013**
Ours	0.013	0.040

- Ours is 2–$6\times$ below LinNash with width constant 1 and 9–$18\times$ below LinNash with the paper's constant (the naive transfer: 7–$15\times$ and 29–$46\times$).
- Phase I lengths in (a): LinNash 13,194, naive 2,850, ours 10,264, FairLinUCB never stops ($= T$).

Results Across p

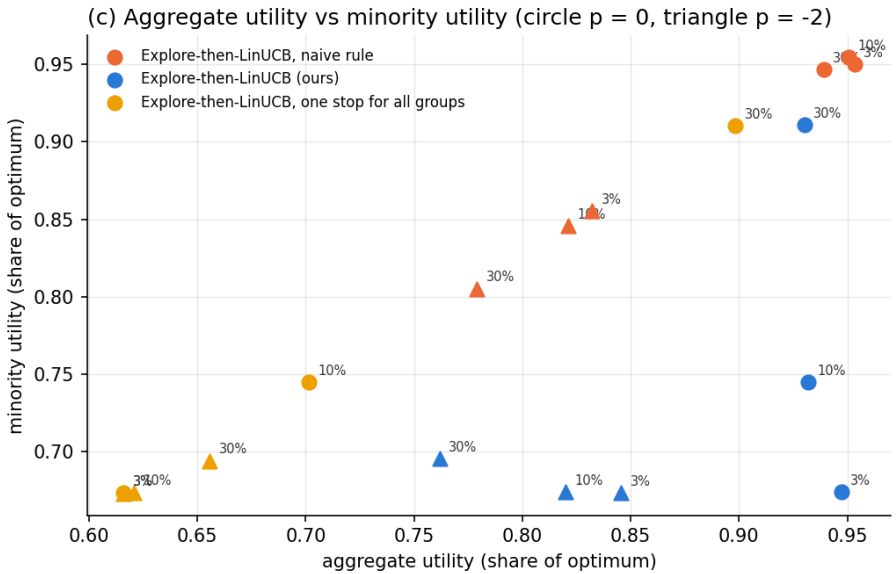


Rare-direction instance (b), regret at T:

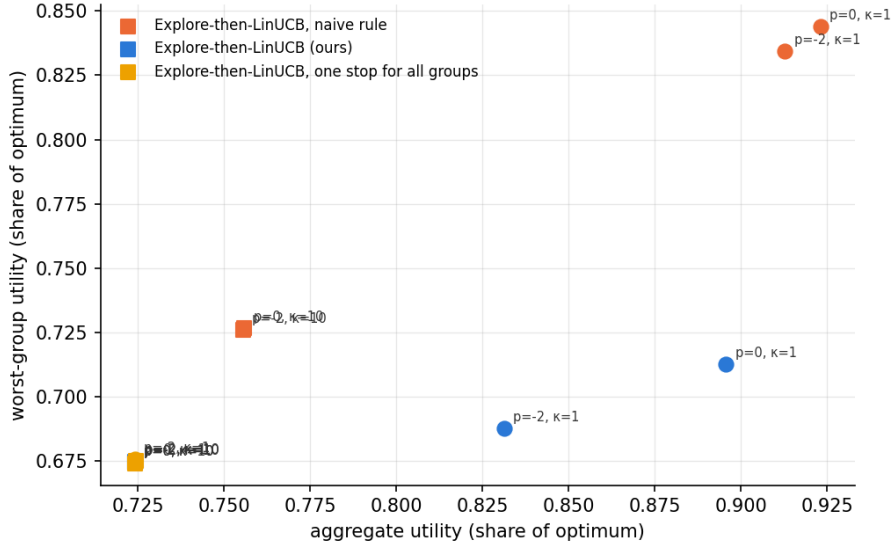
	p=1	p=−2	p=−5
LinNash	0.332	0.666	0.871
LinNash, const. 1	0.084	0.123	**0.194**
FairLinUCB	0.408	0.414	0.420
Naive	**0.010**	**0.074**	0.327
Ours	0.031	0.195	0.412

- For $p \geq -1$: naive $<$ ours $<$ LinNash (const. 1) $<$ LinNash (paper const.), FairLinUCB.
- For $p \leq -2$ the p^2-scaled rule keeps the explore-then variants in Phase I longer; LinNash with a small constant (exploration independent of p) wins at $p = -5$.

Who Pays for Exploration? Minority Groups



two groups



MovieLens-1M (circle κ=1, square κ=10)

- Certifying a minority to the same precision takes about $1/\text{share}$ times more rounds. Per-group stopping: the majority exploits early (utility 0.955) but a 3% minority sits at 0.67; one global stop makes everyone pay (aggregate 0.62).
- MovieLens ($p = 0$, $\kappa = 1$): worst occupation 0.844 (naive) vs. 0.713 (ours). With $\kappa = 10$, $d = 180$, nothing leaves Phase I in $5 \cdot 10^4$ rounds.

Guarantees and Assumptions

An analysis needs (as in [6, 5]): $\langle x, \theta^* \rangle \geq 0$, σ-sub-Gaussian rewards, a valid confidence set (θ^* in the ellipsoid) and a known T.

What our rule certifies (sketch on the good event, not a theorem)

If it fires on a context, $\mathrm{LCB}^* \geq \mu^\star/(1 + 2/\kappa)$, and every item that was plausible at the stop and is later chosen by LinUCB *in that context* has $\mu_a \geq \mu^\star\big(1 - \frac{2}{\kappa|p_a|}\big)$ ($\geq 0.8\,\mu^\star$ for $\kappa = 10$, $p_a = 1$).

Missing, compared with FairLinBandit:

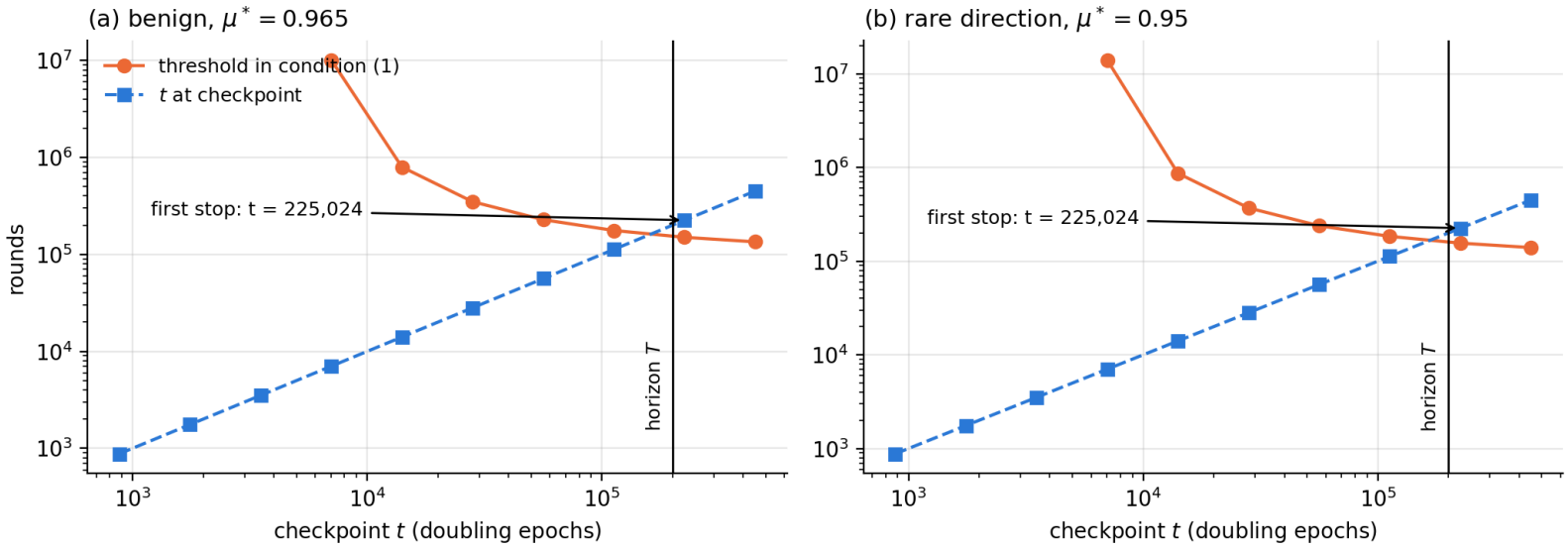
- a bound on τ (they prove $\tau = \Theta(d^2/\mu^{\star 2})$; ours grows like $1/\text{share}$ for minorities);
- an arm-independent width bound over all future contexts (they use G-optimality, $\max_x \|x\|_{V^{-1}} \leq \sqrt{3d/\tau}$);
- a Phase I welfare bound relative to r_t^* (they get $\mu^\star/(2(d+1))$ from the John ellipsoid).

Radius: ours drops the $d\log(1 + t/d)$ term of [1] (otherwise every algorithm explores for the whole horizon at our T), so $\mathrm{LCB}/\mathrm{UCB}$ are *not* proven bounds for $d > 1$.

Where It Fails

- **Strong fairness ($p < -1$).** Stopping scales with p^2: Phase I is $38{,}505$ rounds at $p = -2$ in (a), and the rule never fires within T at $p = -5$ (regret 0.216 in (a), 0.412 in (b)); LinNash with constant 1 reaches 0.095 and 0.194.
- **Longer exploration than needed.** With a LinUCB Phase II the *naive* transfer is better for every $p \geq -1$: ours explores $10{,}264$ vs. $2{,}850$ rounds in (a), $15{,}584$ vs. $5{,}220$ in (b).
- **Minority cost** of per-group stopping (0.67 utility at 3% share).
- **High dimension.** $\kappa = 10$, $d = 180$ never stops in $5 \cdot 10^4$ rounds; $\kappa = 1$ is a heuristic, not the paper's constant.
- **Scope.** LinNash and FairLinUCB need a fixed arm set, so the contextual experiments have no such baseline.

Lower Regret, Not a Better Algorithm



FairLinUCB's stopping condition at its doubling checkpoints, with $\hat{\mu}_{\max} = \mu^\star$ (best case): first pass $t = 225{,}024 > T = 2 \cdot 10^5$.

- In (a), (b) FairLinUCB **never leaves Phase I** ($\tau = T$): its flat 0.21 / 0.41 is the regret of Phase I sampling; the constants (48, 900, $d^2 = 36$) put its first stop just beyond our horizon.
- Ours stops at $\sim 10^4$ because $\kappa|p_a|\beta\|x\|$ uses much smaller constants and a radius without the d term: *better finite-T regret, but the proof is given up.*
- FairLinBandit has $O(d\sqrt{\log T/T})$, order-optimal in d; ours has no bound and addresses a different (contextual) problem.

Conclusion and Future Work

- **Welfarist-UCB.** Adaptive uniform exploration + plain UCB gives near-optimal Nash and p-mean regret for sub-Gaussian rewards. We reproduced its trends: 0.062 vs. 0.225 Nash regret against NCB, large wins over Explore-then-UCB, exploration growing with $|p|$ for $p < -1$.
- **Contextual extension.** Plausible-set stopping rule, design-based exploration, LinUCB Phase II. At $p = 0$, $T = 2 \cdot 10^5$: 2–6$\times$ below LinNash (width const. 1), 9–18$\times$ below LinNash (paper const.); the naive transfer is better still.
- **Honest scope.** Fixed-arm p-mean regret already has an optimal solution, FairLinBandit [5]; ours is empirical, fails for $p \leq -2$ at our horizon, and makes minorities pay for exploration.
- **Future work:** bound τ and give uniform-width / Phase I welfare guarantees for changing contexts; compare with FairLinUCB at $T \gg d^2/\mu^{\star 2}$ or tuned constants; balance exploration cost across groups.

Thank You!

References I

[1] Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. "Improved Algorithms for Linear Stochastic Bandits". In: *NeurIPS*. 2011.

[2] Siddharth Barman et al. "Fairness and Welfare Quantification for Regret in Multi-Armed Bandits". In: *AAAI*. 2023.

[3] Anand Krishna et al. "p-Mean Regret for Stochastic Bandits". In: *AAAI*. 2025.

[4] Hervé Moulin. *Fair Division and Collective Welfare*. MIT Press, 2004.

[5] Dhruv Sarkar, Nishant Pandey, and Sayak Ray Chowdhury. *Improved Algorithms for Nash Welfare in Linear Bandits*. Accepted at ICML. 2026. arXiv: 2601.22969 [cs.LG].

[6] Dhruv Sarkar, Nishant Pandey, and Sayak Ray Chowdhury. *Revisiting Social Welfare in Bandits: UCB is (Nearly) All You Need*. 2025. arXiv: 2510.21312 [cs.LG].

References II

[7] Ayush Sawarni, Soumyabrata Pal, and Siddharth Barman. "Nash Regret Guarantees for Linear Bandits". In: *NeurIPS*. 2023.